

The Non-Associativity Selector: A Complete Classification of $\text{Im}(\mathbb{O})$'s Generalized Inönü–Wigner Contractions, and Why 1+3+3 Is Forced

Harald Kjøge Rønning

[OPEN 3]'s first stage, closed. Every admissible grading of the seven imaginary octonion directions — any partition, any number of grade levels, any positive real weights — contracts to either a Lie algebra or the R-flux 1+3+3 pattern, and nothing else. The census that found this (484 realizable patterns, 14 orbits under the Fano symmetry) is independently reproducible and reported in full; the theorem that decides it does not depend on the census at all. Base hash of the verifying code available on request.

Revision 2. Restructured for the working corpus: the error ledger removed (preserved separately in project development notes), and the open-items ledger consolidated — shared items now cite the central registry by ID rather than restating them.

1. Setup: gradings, validity, patterns

[DEF, from Keystone §2] A *generalized Inönü–Wigner contraction* of $\text{Im}(\mathbb{O})$ along a grading $g : \{1, \dots, 7\} \rightarrow \mathbb{R}_{>0}$ (one weight per imaginary unit) rescales each unit by $\lambda^{g(i)}$ and keeps, for each bracket $[e_i, e_j]$, only the component landing exactly at grade $g(i) + g(j)$ as $\lambda \rightarrow 0$: lower-grade components vanish in the limit; a *valid* grading is one where no higher-grade component exists to begin with, since such a component would diverge rather than vanish. Keystone's own 1+3+3 case is the instance where g takes exactly three values, with one unit at the top value playing the central role.

[FACT, structural] Every octonionic bracket $[e_i, e_j]$, $i \neq j$, is single-component: the imaginary octonion units, as points of the Fano plane under the standard multiplication table, satisfy — for every pair i, j — $[e_i, e_j] = \pm 2e_{k(i,j)}$ for a unique third point $k(i, j)$, the third point of the Fano line through i, j . In the basis convention used throughout this document, indices are XOR-composed: $k(i, j) = i \oplus j$ (binary XOR on $\{1, \dots, 7\}$), which fixes the product-index map completely — and that map is all any definition or computation in this document uses; the signs $\pm$ in $[e_i, e_j] = \pm 2e_{k(i,j)}$ vary across octonion sign conventions sharing this same index map, but never enter a claim made here, since every result below depends only on which basis element a bracket lands on, never on the coefficient it lands with. This is the sole structural fact the entire classification below is built from: validity reduces to the single inequality $g(k(i, j)) \leq g(i) + g(j)$ for each of the 21 pairs, an equality marking a *tight* pair — the pair's bracket survives the contraction — and a strict inequality marking one that vanishes.

[DEF] The *pattern* of a valid grading g is the assignment, for each of the 21 pairs, of *tight* or *vanishes*. Two gradings with the same pattern induce the identical contracted algebra: the surviving brackets are exactly the tight pairs, each with its original (uncontracted) structure constant. The classification problem [OPEN 3]'s decidable first stage asks about — is 1+3+3 forced, or one of a landscape? — is a question about patterns, not about the (in principle infinite) space of gradings that induce them.

[REMARK, on real weights] Nothing below uses integrality. The argument is order-and-addition throughout: an inequality $g(k) \leq g(i) + g(j)$, an equality forcing $g(k) = g(i) + g(j)$ exactly, a chain of such equalities forcing a contradiction. Every claim in §§2–4 holds for $g : \{1, \dots, 7\} \rightarrow \mathbb{R}_{>0}$

without modification; integer witnesses, used freely in §5’s census for enumerability, are a convenience there, never a hypothesis of the theorems.

2. Theorem A: no chain means Lie, unconditionally

[THM, proven] Let g be a valid grading whose pattern contains no *depth-2 chain* — no tight pair $(i, j) \rightarrow k$ whose target k is itself an input to some other tight pair. Then the contracted algebra is 2-step nilpotent: $[[x, y], z] = 0$ identically, for all x, y, z . In particular it is Lie (trivially Jacobi) and Malcev — trivially, since $J \equiv 0$ identically for any Lie algebra (Jacobi holding for every input, not merely fixed ones) forces both sides of the Malcev identity, *in its Jacobiator form* $J(x, y, [x, z]) = [J(x, y, z), x]$, to vanish outright: the left side is the *full* three-term Jacobiator $J(x, y, [x, z])$, not any single bracket term inside it, and $J \equiv 0$ kills it regardless of what $[x, z]$ is; the right side $[J(x, y, z), x] = [0, x]$ vanishes the same way. This holds for every Lie algebra, not merely this document’s 2-step nilpotent case — not vacuously in the technical sense of a hypothesis that never fires, but as a direct computation. (Equivalently, in the expanded Sage form $[[x, y], [x, z]] = [[[[x, y], z], x] + [[[[y, z], x], x] + [[[[z, x], x], y]$ used in at least one of this document’s numerical Malcev checks in §6, both sides are generically *nonzero but equal* for a Lie algebra — the two standard forms of the identity are interchangeable, and the vanishing description above belongs to the Jacobiator form specifically, not to every equivalent way of writing Malcev.)

Proof. “No chain” means precisely that every target of every tight pair is *input-free* — it never appears as one of the two arguments of any tight pair. For any tight pair $(x, y) \rightarrow k$, $[x, y]$ is (a multiple of) e_k ; since k is input-free, $[e_k, z]$ is never tight for any z , hence vanishes in the contraction, for every z . So $[[x, y], z] = 0$ for every tight pair (x, y) and every z . For a non-tight pair (x, y) , $[x, y] = 0$ directly, so $[[x, y], z] = [0, z] = 0$ trivially. Either way, $[[x, y], z] = 0$ for all x, y, z . ■

3. Theorem B: a chain forces the R-flux 1+3+3 pattern, exactly

[THM, proven] Let g be a valid grading whose pattern contains a depth-2 chain: tight pairs $(i, j) \rightarrow k$ and $(k, m) \rightarrow n$, for some $i, j, k, m, n \in \{1, \dots, 7\}$. Then the pattern is exactly the six-tight-pair R-flux 1+3+3 configuration: $X = \{i, j, m\}$ (input-only), $P = \{k, p, q\}$ for the two remaining points p, q (each both an input and a target), $W = n$ (target-only, the central generator).

Proof, in eight steps.

(1) The third line through k . $L_1 = \{i, j, k\}$ and $L_2 = \{k, m, n\}$ are distinct Fano lines sharing k (a second tight pair on L_1 is excluded by the at-most-one-tight-pair-per-line fact, itself immediate from the algebra: two tight pairs on one line force a grade to zero). Every point of the Fano plane lies on exactly three lines, and the three lines through a fixed point partition the remaining six points into three pairs (the Fano matching lemma, established independently in a companion paper on four-level shapes). L_1, L_2 account for two of k ’s three lines, pairing $\{i, j\}$ and $\{m, n\}$; the third line through k , L_3 , pairs the two points not yet accounted for, p, q — so $L_3 = \{k, p, q\}$, and $\{i, j, k, m, n, p, q\}$ exhausts all seven points.

(2) n ’s other two lines. By the same matching lemma applied to n : L_2 accounts for one of n ’s

three lines (pairing k, m); the other two partition the remaining four points $\{i, j, p, q\}$ into two pairs. $\{n, i, j\}$ cannot be one of them, since i, j already determine L_1 uniquely (two points determine a unique line), and $L_1 \not\supset n$. So n 's other two lines pair i with one of $\{p, q\}$ and j with the other; WLOG (relabeling p, q) these are $\{n, i, p\}$ and $\{n, j, q\}$.

(3) m 's forced lines. By the matching lemma for m : L_2 accounts for one line (pairing k, n); the other two partition $\{i, j, p, q\}$. m 's line through i cannot contain j (i, j already determine $L_1 \not\supset m$), cannot contain k or n (already m 's partners via L_2), and cannot contain p (step 2 already fixed i, p on the line $\{n, i, p\}$, and two points determine a unique line). The only remaining candidate is q : m 's line through i is forced to be $\{m, i, q\}$. Symmetrically, m 's line through j is forced to be $\{m, j, p\}$.

(4) The grade equations. Write $a = g(i)$, $b = g(j)$, $d = g(m)$. The two given tight pairs give $g(k) = a + b$ and $g(n) = g(k) + g(m) = a + b + d$. Validity on the pair $(i, p) \rightarrow n$ (from $\{n, i, p\}$, step 2) gives $g(n) \leq g(i) + g(p)$, i.e. $g(p) \geq b + d$. Validity on $(j, q) \rightarrow n$ (from $\{n, j, q\}$) gives $g(q) \geq a + d$.

(5) The first squeeze. Validity on the pair $(m, i) \rightarrow q$ (from $\{m, i, q\}$, step 3) gives $g(q) \leq g(m) + g(i) = a + d$. Combined with step 4's $g(q) \geq a + d$: $g(q) = a + d$ exactly. The pair $(m, i) \rightarrow q$ is therefore tight, forced — an upper bound from one line meeting a lower bound from another, with nothing free left over.

(6) The second squeeze, symmetric. Validity on $(m, j) \rightarrow p$ gives $g(p) \leq b + d$; combined with step 4's $g(p) \geq b + d$, $g(p) = b + d$ exactly, and $(m, j) \rightarrow p$ is tight.

(7) Two more tight pairs, for free. $g(i) + g(p) = a + (b + d) = a + b + d = g(n)$, so $(i, p) \rightarrow n$ is tight (not merely valid — exactly tight). Symmetrically $g(j) + g(q) = b + (a + d) = a + b + d = g(n)$, so $(j, q) \rightarrow n$ is tight.

(8) L_3 carries no tight pair. $L_3 = \{k, p, q\}$ has three candidate pairs. $(k, p) \rightarrow q$ tight would need $g(q) = g(k) + g(p) = (a + b) + (b + d)$, but $g(q) = a + d$ (step 5): equating forces $2b = 0$, impossible for positive b . $(k, q) \rightarrow p$ tight would force $2a = 0$; $(p, q) \rightarrow k$ tight would force $2d = 0$. All three are impossible, so L_3 is untight.

Collecting: six tight pairs total — $(i, j) \rightarrow k$, $(k, m) \rightarrow n$, $(m, i) \rightarrow q$, $(m, j) \rightarrow p$, $(i, p) \rightarrow n$, $(j, q) \rightarrow n$ — covering six of the seven lines, with L_3 the sole untight line. Since each line carries at most one tight pair, these six named pairs saturate their six lines exactly, and L_3 carries none, so the pattern is exactly these six — no seventh tight pair is possible anywhere, on any of the seven lines. i, j, m appear only as inputs, never targets; n appears only as a target, never an input; k, p, q each appear as both. This is exactly the R-flux 1+3+3 pattern: $X = \{i, j, m\}$, $P = \{k, p, q\} = L_3$, $W = n$. ■

[REMARK, the closed form] The proof is constructive: for *any* positive reals a, b, d , the grading $g(i)=a$, $g(j)=b$, $g(m)=d$, $g(k)=a+b$, $g(p)=b+d$, $g(q)=a+d$, $g(n)=a+b+d$ is valid and realizes the pattern directly — no search required. Verified exactly, symbolically, against every algebraic step above; verified again on a concrete instance traced through the actual Fano incidence structure ($i, j, k, m, n, p, q = 1, 2, 3, 4, 7, 6, 5$ in the standard labeling); and verified once more with genuinely irrational weights ($a, b, d = \pi, \sqrt{2}, e$), confirming no integrality is used anywhere in the construction.

4. The corollary, and where the free input finally sits

[COR, the classification, unconditional] Every valid grading of $\text{Im}(\mathbb{O})$ — any partition, any number of grade levels, any positive real weights, enumerated or not — realizes either a pattern with no depth-2 chain, hence contracts to a 2-step nilpotent Lie algebra (Theorem A), or realizes the R-flux 1+3+3 pattern, hence contracts to the R-flux algebra (Theorem B). No third outcome exists, and no enumeration is needed to establish this: Theorem A’s hypothesis and Theorem B’s hypothesis are exhaustive and mutually exclusive by definition — a pattern either contains a depth-2 chain or it does not. The dichotomy is substantive, not a tautology dressed as a theorem, only because the R-flux algebra genuinely fails Jacobi: the Jacobiator itself, $J(X^a, X^b, X^c) \propto \epsilon^{abc}W$, is Günaydin–Lüst–Malek’s own R-flux signature (the bracket $\{X, X\} \propto P$, from the corpus’s own companion paper on the necessary and sufficient R-flux condition, is a necessary ingredient of this structure but not itself the Jacobi-failure that makes the dichotomy interesting). Verified directly across all 35 basis triples: every nonzero value of the Jacobiator lies along W , and no other triple type produces one — confirming the signature exactly, not merely its plausibility.

[COR, restated as the terminal relocation] [OPEN 3]’s free input, tracked across this project’s own ledger from “why three coordinate and momentum directions” to “why canonical pairing at all,” relocates one final time and stops: the single hypothesis this classification needs is *that the contraction retain any non-associative content whatsoever*. Demand that, and the 1+3+3 R-flux pattern is forced outright — the central generator, the canonical pairing, and the three-dimensionality of each block all arrive as consequences, not further inputs. That hypothesis is not a further unexplained choice: it is this project’s founding motivation, the entire reason the construction begins from $\mathbb{O}$ rather than an associative algebra. The relocation chain Keystone’s own [PROP] predicted — from an arbitrary numerical choice toward a named structural principle — bottoms out here.

5. The census: the full landscape, independently reproducible

[SCOPE] This section is historically honest but not load-bearing. It is the route by which Theorem B was found — the structural argument in §3 was only visible once the pattern table existed to look at — and it remains independently valuable as the complete picture. But §§2–4 do not depend on it: the corollary holds regardless of whether the count below is exactly right.

[FACT, computed] Restricting integer grades to $\{1, \dots, 9\}$ finds **484 realizable valid patterns**, with the count stabilizing already at $\{1, \dots, 4\}$ — six consecutive grade bounds (4 through 9) all returning the identical count, with no new pattern appearing as the bound more than doubles. This stabilization is empirical, not yet a proof that no larger grade could add a 485th pattern; §5’s closing paragraph states the residual gap precisely. These form **14 orbits** under the group of permutations of the seven imaginary units preserving the Fano line structure — the collineation group $\text{PGL}(3, 2)$, order **168**, confirmed both by direct enumeration (168 of the 5040 permutations of seven points preserve the line structure) and by its standard identification. Orbit computation agrees exactly across all three independent methods used (canonical-form deduplication, Burnside counting, and direct group-action closure via union-find — detailed in §6), orbit for orbit.

[TABLE, the 14 orbits, generated directly from code at draft time]

orbit size	tight pairs	has depth-2 chain	witness grading ($g_1, \dots, g_7$)
1	0	no	(1, 1, 1, 1, 1, 1, 1) — trivial
7	3	no	(1, 1, 1, 1, 1, 1, 2) — Heisenberg / Orbit A
7	6	no	(1, 1, 2, 1, 2, 2, 1) — quaternionic- Heisenberg (below)
21	1	no	(1, 1, 2, 2, 2, 2, 2)
21	2	no	(2, 2, 2, 2, 2, 3, 4)
21	4	no	(1, 1, 1, 1, 1, 2, 2)
28	3	no	(1, 1, 1, 1, 2, 2, 2)
28	3	no	(1, 1, 2, 1, 2, 2, 2)
28	3	no	(1, 2, 3, 2, 3, 3, 2)
28	6	yes	(1, 1, 2, 1, 2, 2, 3) — R-flux / Orbit B
42	5	no	(1, 1, 2, 2, 3, 3, 2)
84	2	no	(1, 2, 2, 2, 3, 2, 3)
84	3	no	(1, 1, 2, 2, 2, 2, 3)
84	4	no	(1, 2, 3, 2, 3, 4, 3)

Sizes sum to 484; every size divides 168; exactly one orbit carries a depth-2 chain, matching Theorem B’s forcing exactly, and none of the other thirteen does, matching Theorem A.

[FACT, checkable footnote] The size-7, tight-6 orbit — no depth-2 chain, so Theorem A applies and it is Lie despite its six surviving brackets — has three central targets, each receiving a pairing of rank 4 over the same four generators. This is the seven-dimensional quaternionic Heisenberg algebra, $\mathbb{R}^4 \oplus \text{Im } \mathbb{H}$: the landscape’s most structurally elaborate also-ran, previously unmet in this corpus, now named.

[COR, the two-part conjecture, resolved at pattern level] Keystone’s [PROP] refinement asked: is every viable grading Heisenberg-shaped, and is every Heisenberg-shaped grading exactly 1+3+3? Scanning the fourteen structural keys directly: exactly two orbits have exactly one central unit that is also a target with full-rank pairing — Heisenberg and R-flux, and no other. Combined with §2–3’s classification, the second half holds in a stronger form than conjectured: criteria beyond “retains non-associative content” were never needed for uniqueness.

[SCOPE, the remaining formality] “Exactly 484, no more” — the census’s own completeness — was screened by two independent methods (an LP feasibility pass over the 4^7 per-line reduction, and exhaustive integer search up to grade 9, with the count stabilizing from grade 4). A targeted exhaustive sweep of all $6^7 = 279,936$ gradings does two separate jobs, worth stating as two: it corroborates the census (no 485th pattern appears among these 279,936 gradings), and, independently, it corroborates

Theorem B directly — of the 560 chained instances found, every one matches the R-flux signature with zero exceptions, the same conclusion Theorem B reaches without any sweep at all. Exact-arithmetic infeasibility certificates for the mechanically-rejected candidates remain on the ledger as a bounded, small formality — explicitly not load-bearing for §4’s corollary, which does not quantify over the census at all.

6. Method: two independent implementations, a commit-reveal protocol

[FACT] Every quantitative claim above and in this section was reproduced independently by two separately-built implementations (the author’s, and a fresh-context reviewer’s), built from source-grounded definitions rather than shared code, cross-validated at each stage before proceeding to the next: the 140/56 count on Keystone’s own 1+3+3 shape, the 420/0 and 140/0 counts on two structurally-impossible four-level shapes (sizes 3, 2, 1, 1 at grades 1, 2, 3, 4; sizes 3, 3, 1 at grades 1, 2, 4), and the 630/84 count on the four-level 2+2+2+1 shape — all four drawn from, or generalizing, this project’s own companion paper on four-level shapes; the 484-pattern, 14-orbit landscape, confirmed via three different orbit-computation methods (canonical-form deduplication, Burnside counting, direct union-find on the group action); the four stage-2 criteria (Malcev on random exact-integer triples, validated against $\text{Im}(\mathbb{O})$ passing and $\text{Im}(\mathbb{S})$ failing; the Jacobiator on all 35 basis triples; centrality; strict Heisenberg rank), calibrated against the three corpus-known shapes before use on any novel orbit.

[FACT, the commit-reveal protocol] To rule out anchoring, stage-2 verdicts for the eleven previously-unstudied orbits were computed independently by both sides before comparison: one side published calibration rows and a SHA-256 commitment over its full table, revealing only after the other side’s independent table was complete. A structural key collision, caught during this process (two distinct orbits sharing identical (size, tight-count, target-profile) identifiers), voided the first commitment; the corrected commitment and the independently-reproduced table are both on record:

- Voided commitment (superseded due to the key collision, reported fully rather than truncated or silently replaced):

e3fe27acddfeed1e2ca6ff3d16adcc1961901d56dbdca8013b2ccba8c74c2609

- Superseding commitment:

9a62cee492aa9ac363d59d7a21f0c89d715f88b89a1c086b130363bb484fbb4f

- Independently reproduced table’s hash:

f16e78b56abdca96a5fd5b3133ffcf5d6ede11a913d0c899bb9035fb4e5b3ed0

Diff on reveal: zero mismatches across all fourteen rows and every shared column.

7. Scope

[SCOPE] Three items, stated precisely.

(i) The classification concerns *single-grading* contractions of $\text{Im}(\mathbb{O})$ specifically. Keystone’s own construction applies a second Inönü–Wigner contraction after the first (§3 of that document, the dilation and the $\mathfrak{so}(3)$ remnant); this theorem says nothing about iterated contractions, and “unique non-associativity-retaining pattern” should not be read as “unique route to anything downstream.” It is not, and the corollary does not need it to be.

(ii) Keystone’s own [SCOPE], restated verbatim rather than left to drift: “Every contraction in this chain, including the enumeration proposed above, is a construction device — a tool for building the operational algebra from $\text{Im}(\mathbb{O})$ — not an asserted physical process the seed is claimed to undergo.” This document’s theorem is a statement about which gradings are structurally available, not a claim about physical dynamics.

(iii) Criteria 3 (central generator) and 4 (strict Heisenberg shape, full-rank pairing) — defined and computed in §5 for completeness and cross-validation — are not needed for the uniqueness result. Criterion 2 (surviving non-associative content) alone decides it, per §§2–4; §5’s fuller stage-2 table is corroboration and context, not part of the proof.

[LEDGER] Open: the exact-arithmetic infeasibility certificates for the census’s mechanically-rejected candidates, a bounded formality not required by §4’s corollary but owed to the census’s own completeness claim; whether the depth-2-chain classification technique generalizes to a comparably clean, enumeration-free proof of the exact orbit *count* (not merely the chain/no-chain dichotomy this document needs); Registry REG-3 (whether [OPEN 1]’s “single source of dynamics” principle and this document’s single hypothesis are secretly one requirement).

References

- Companion document: *The Keystone: From $(\mathbb{S}, \tau)$ to the Injective Type III_1 Factor* (§§1–2, the grading ansatz, [OPEN 3], [TARGET], [PROP]).
 - Companion paper: *The X/P Asymmetry Was Never a Law: Joint Survival at Four Grade Levels* (the Fano matching lemma, the $2+2+2+1$ shape’s own validity census).
 - Companion paper: *Which Contractions of the Octonions Give Genuine R-Flux: A Necessary and Sufficient Condition* (the Orbit A / Orbit B distinction this document’s census reproduces and generalizes).
 - Günaydin, M., Lüst, D. & Malek, E., “Non-associativity in non-geometric string and M-theory backgrounds, the algebra of octonions, and missing momentum modes,” arXiv:1607.06474.
 - Weimar-Woods, E., “Contractions, generalized Inönü–Wigner contractions and deformations of finite-dimensional Lie algebras,” *Rev. Math. Phys.* 12 (2000), 1505–1529.
-

Appendix A: the exact serialized stage-2 table

[FACT, reproducibility] §6 publishes three commitment hashes but not the table they bind. This appendix closes that gap: the verbatim string below, hashed with SHA-256, reproduces the “independently reproduced table’s hash” reported in §6 exactly — specifically the third of

the three hashes there (f16e78b5...), the one committing to table *content*; the other two (the voided e3fe27ac... and the superseding 9a62cee4...) bind a pre-reveal sealing process over working logs this appendix does not reproduce, and are not what recomputing the block below checks. Format: fourteen rows, one per orbit, sorted by (size, tight-count); each row is size,tight,malcev,non-Lie,criterion-2,criterion-3,criterion-4-strict,rank (the last field is the σ -rank where strict criterion 4 applies) with Python boolean literals (True/False) and None where strict criterion 4 does not apply (no unique central target); rows are semicolon-terminated with no separating whitespace. This is the exact string that was hashed — reproduced character-for-character, not reformatted — and recomputing its SHA-256 against the published commitment is a two-line check for anyone with a hashing utility.

```
1,0,True,False,False,False,False,None;
7,3,True,False,False,True,True,6;
7,6,True,False,False,True,False,None;
21,1,True,False,False,True,False,None;
21,2,True,False,False,True,False,None;
21,4,True,False,False,True,False,None;
28,3,True,False,False,True,False,None;
28,3,True,False,False,True,False,None;
28,3,True,False,False,True,False,None;
28,6,True,True,True,True,True,6;
42,5,True,False,False,True,False,None;
84,2,True,False,False,True,False,None;
84,3,True,False,False,True,False,None;
84,4,True,False,False,True,False,None;
```

Fourteen lines, one row per orbit, displayed this way purely for page width — the line breaks above are *not* part of the hashed string. Concatenating all fourteen lines with no separator (each already ends in the semicolon that terminates its row) reproduces the exact 520-character string that was hashed.

SHA-256 of the block above: f16e78b56abdca96a5fd5b3133ffcf5d6ede11a913d0c899bb9035fb4e5b3ed0 — matching §6’s published commitment exactly.

[TABLE, the same data, formatted for reading] Columns: orbit size; tight pairs; Malcev; non-Lie (nonzero Jacobiator); criterion 2 (Malcev $\wedge$ non-Lie); criterion 3 (a central target exists); criterion 4, strict (exactly one central unit, itself a target, full-rank 6 pairing); σ -rank where criterion 4 applies; witness grading (from §5’s table, repeated here so this appendix is self-contained).

size	tight	Malcev	non-Lie	crit 2	crit 3	crit 4	rank	witness
1	0	T	F	F	F	F	—	(1, 1, 1, 1, 1, 1, 1)
7	3	T	F	F	T	T	6	(1, 1, 1, 1, 1, 1, 2)
7	6	T	F	F	T	F	—	(1, 1, 2, 1, 2, 2, 1)
21	1	T	F	F	T	F	—	(1, 1, 2, 2, 2, 2, 2)
21	2	T	F	F	T	F	—	(2, 2, 2, 2, 2, 3, 4)
21	4	T	F	F	T	F	—	(1, 1, 1, 1, 1, 2, 2)
28	3	T	F	F	T	F	—	(1, 1, 1, 1, 2, 2, 2)
28	3	T	F	F	T	F	—	(1, 1, 2, 1, 2, 2, 2)

size	tight	Malcev	non-Lie	crit 2	crit 3	crit 4	rank	witness
28	3	T	F	F	T	F	—	(1, 2, 3, 2, 3, 3, 2)
28	6	T	T	T	T	T	6	(1, 1, 2, 1, 2, 2, 3)
42	5	T	F	F	T	F	—	(1, 1, 2, 2, 3, 3, 2)
84	2	T	F	F	T	F	—	(1, 2, 2, 2, 3, 2, 3)
84	3	T	F	F	T	F	—	(1, 1, 2, 2, 2, 2, 3)
84	4	T	F	F	T	F	—	(1, 2, 3, 2, 3, 4, 3)

The full table carries the invariants §5’s simplified table omits — including the σ -rank data behind the quaternionic-Heisenberg identification (the (7, 6) row: three central targets, rank 4 each, discussed in §5’s checkable footnote).

Appendix B: the seven Fano lines, for hand-checking

[FACT] Under this document’s XOR convention ($k(i, j) = i \oplus j$, §1), the seven Fano lines are:

line	check
$\{1, 2, 3\}$	$1 \oplus 2 = 3$
$\{1, 4, 5\}$	$1 \oplus 4 = 5$
$\{1, 6, 7\}$	$1 \oplus 7 = 6$
$\{2, 4, 6\}$	$2 \oplus 4 = 6$
$\{2, 5, 7\}$	$2 \oplus 5 = 7$
$\{3, 4, 7\}$	$3 \oplus 4 = 7$
$\{3, 5, 6\}$	$3 \oplus 6 = 5$

Displayed once, this makes Theorem B’s eight steps checkable entirely on paper: every incidence claim in §3’s proof (which lines pass through which points, which pairs share a line) is directly readable from this table without recomputing $k(i, j)$.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author’s responsibility.